Plastic-PMT High-Voltage Scan: Gain Curves, HV Equalization, and Stability of the SiPM-Wall MIP Calibration

n_TOF EAR2 X17 campaign, run 224466 (July 16, 2026)

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Abstract

During run 224466 the eight plastic-scintillator PMTs were stepped together through 1200 V to 1600 V in nine ∼10-minute, proton-gated steps (two interleaved passes on a 50 V grid), while the n_TOF DAQ kept recording. Slicing the run by wall-clock time gives per-voltage amplitude spectra for every PMT. Wall-tagged coincident spectra yield clean per-PMT gain power laws $G \propto V^n$ with $n = 4.8$–6.9, from which we derive a voltage-equalization table (current response spread 2.8×; largest corrections −120 V on C-left and +117 V on D-right). The key systematic check for the MIP program: the SiPM-wall MIP peak obtained through wall–plastic coincidences is stable to ±2.6% (one histogram bin) on all four arms across the entire plastic-HV range — a 3× change in tag rate — and across the SiPM flash-gating change, so the wall MIP calibration is independent of the plastic operating point. The scan also shows no plastic rate plateau up to 1600 V, and a surprisingly small (∼4–6%) effect of the "flash gating" on wall occupancy in the γ-flash window, flagged for hardware follow-up. For the plastic trigger, the discriminator's 10 mV hardware floor keeps 86–97% of the wall-tagged population per PMT at the equalized gains (unconstrained, 4 mV to 5 mV would give ≥99% everywhere); the measured power laws quantify the HV increase that would recover full efficiency (×2.35 gain) once an absolute MIP calibration justifies it.

1 Scan design and data pairing

The scan tool (CAEN card 07) held all eight plastic PMTs at a common voltage per step and advanced after $\approx 10^{15}$ protons on target, logging HV and beam only; the PMT signals themselves are in the regular n_TOF stream (run 224466, 13:04–15:19 local, 2101 bunches). Pass 1 stepped 1600 → 1200 V in 100 V steps with the SiPM γ-flash gating *off*; pass 2 stepped 1550 → 1250 V with gating *on*, so the two passes interleave to a 50 V grid but differ in gating state. Each bunch is assigned to a step through its pick-up-detector timestamp (`psTime`, ns since epoch; scan windows from the HV log). Two extra windows come for free: the pre-scan stretch with all PMTs at 1500 V (gating off) and the post-scan return to the per-channel nominals (A-L 1325, A-R 1275, B-L 1325, all others 1300 V; gating on). Steps collect 84–146 good bunches each (401/308 for the pre/post windows); all observables are normalized to the delivered protons per step, and the standard selection of the MIP analysis is applied (beam-on bunches, hits > 0.1 ms after the γ-flash). The scan layout and the per-step exposure are summarized in Fig. 1.

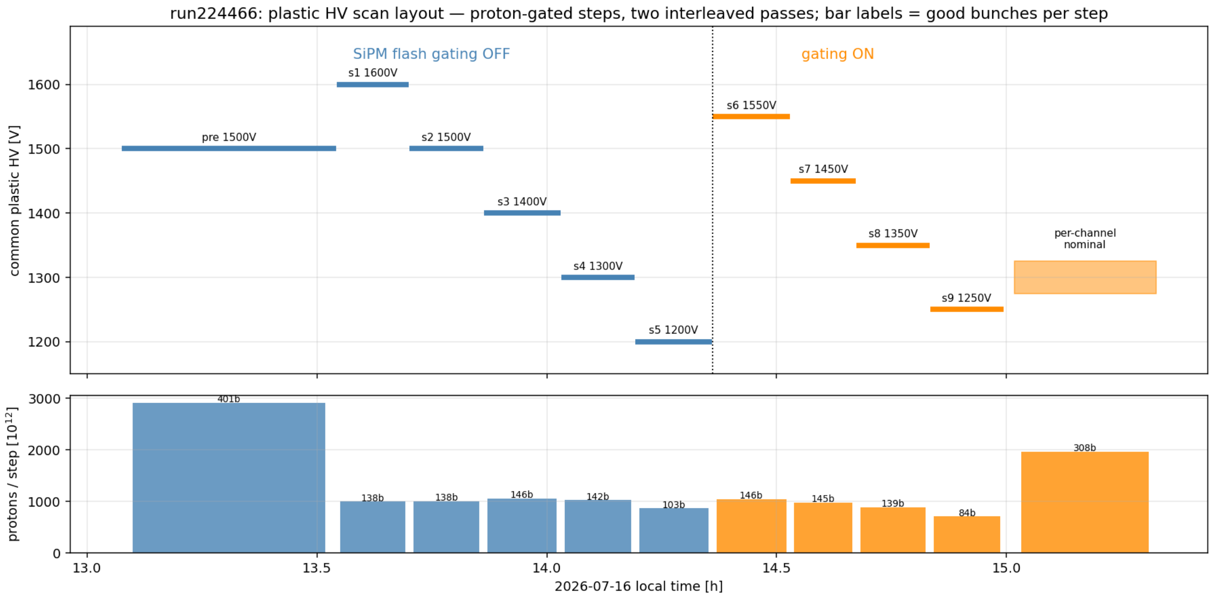

Figure 1: Scan layout: common plastic HV vs. wall-clock time (top; blue = flash-gating off, orange = on, dotted line = changeover) and delivered protons per step (bottom; labels = good bunches per step). The pre-scan 1500 V stretch and the post-scan return to per-channel nominals provide two extra calibration windows.

2 Gain observable: wall-tagged coincident spectra

The *inclusive* plastic hit spectrum is a poor gain observable: the acquisition threshold sits well inside the steeply falling spectrum, so as the gain rises more small pulses enter the sample and the median hardly moves (for three PMTs it even decreases with voltage). Instead we use the wall-tagged sample of the MIP analysis: plastic hits in a ± 8 ns calibrated coincidence with a same-arm wall channel (time offsets re-measured on this run), sideband-subtracted with the $+20$ to $+120$ ns window. The tag is provided by the SiPM wall, whose response is independent of the plastic bias, so the tagged plastic population is a fixed physical reference whose measured amplitude tracks the PMT gain directly.

Figure 2 shows the tagged spectra marching up with voltage at fixed shape (log amplitude axis); Figure 3 shows the same spectra as probability densities on a linear amplitude axis, with an arrow marking each spectrum's median — the single number per (PMT, voltage) that the rest of the analysis uses. The median is chosen over a peak fit because the tagged plastic spectrum has no sharp MIP peak (the plastic sits behind the wall and the tag does not constrain path length in the plastic); once the liquid-scintillator readout is available, a wall$\times$plastic$\times$LS triple coincidence should isolate a true plastic MIP peak and allow an absolute (rather than relative) energy calibration. Figure 4 shows the median of the subtracted spectrum versus the measured bias on log–log axes. Each PMT follows a single power law med $\propto V^n$: the two passes land on one curve (the gating state does not disturb the gain measurement), the pre-scan 1500 V window reproduces the pass-1 1500 V step exactly, and the post-scan nominal points fall on the fits. Fitted indices are listed in Table 1; C-left, with the lowest index ($n = 4.8$), visibly flattens above $\sim$1500 V.

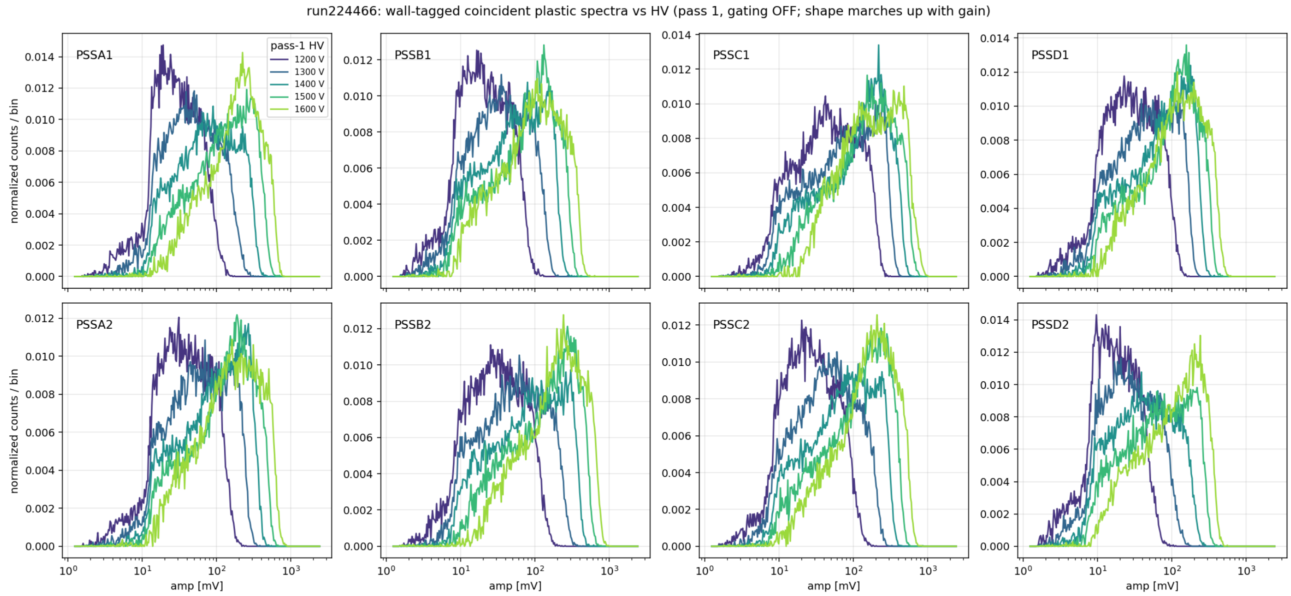

Figure 2: Wall-tagged, sideband-subtracted plastic amplitude spectra for the five pass-1 voltages (gating off), normalized to unit area. The population is fixed by the wall tag; the amplitude scale moves with the PMT gain.

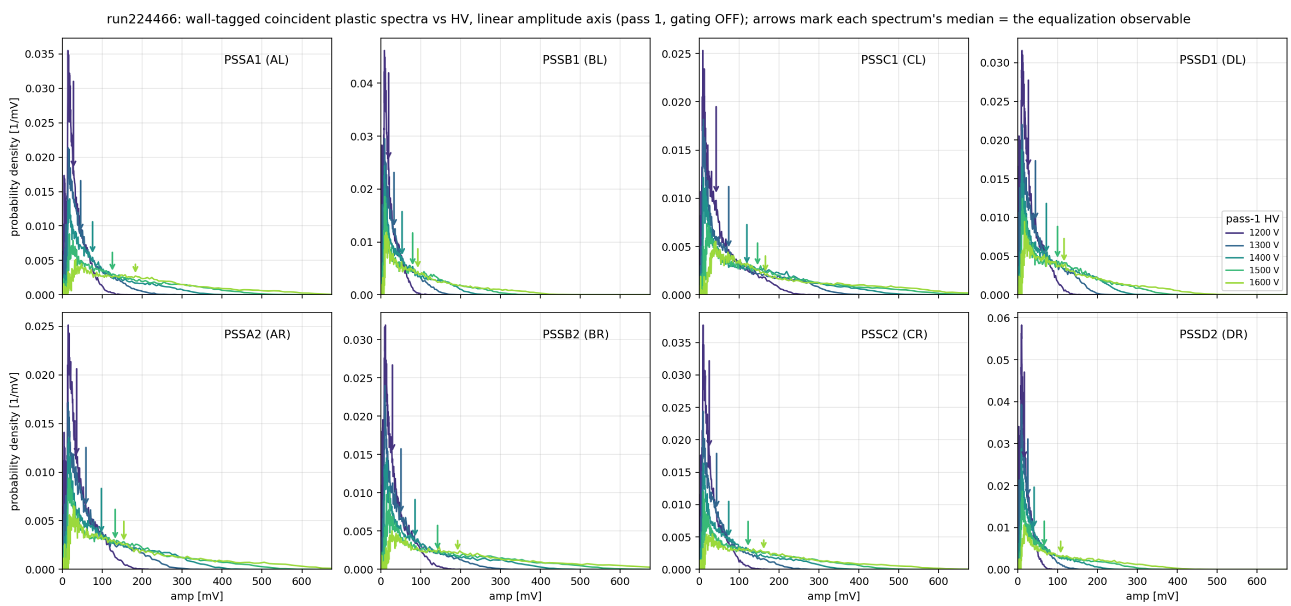

Figure 3: The same tagged spectra as densities on a linear amplitude axis (pass 1), in mV (per-channel ADC→mV factors, all $\approx 0.0307\,\mathrm{mV/count}$). Arrows mark each spectrum's median, the equalization observable. The low-amplitude pile-up against the acquisition threshold, prominent at low voltage, is why the *inclusive* median is not a usable gain proxy.

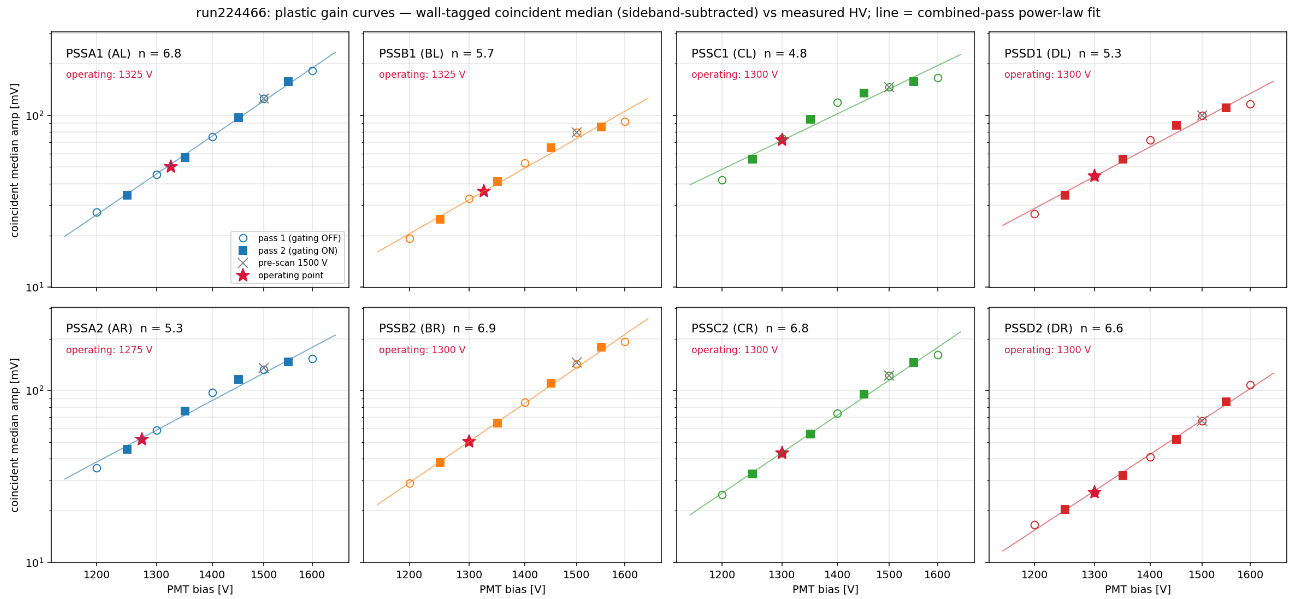

Figure 4: Coincident median amplitude vs. measured bias (log–log), per PMT; the operating voltage is noted in each panel. Open circles: pass 1 (gating off); filled squares: pass 2 (gating on); gray cross: pre-scan 1500 V window; star: post-scan operating point. Lines: combined-pass power-law fits (indices in Table 1).

3 HV equalization

At the current operating voltages the tagged-median response spans a factor 2.8 (C-left highest, D-right lowest). Table 1 gives, per PMT, the voltage that brings its response to a common target using that PMT's own fitted index; Fig. 5 shows the construction — each PMT slides along its own gain curve to the target line. These measured exponents supersede the $G \propto V^7$ assumption used for the provisional suggestions in the run-224404 report.

Choice of the absolute target. The target is simply the *geometric mean of the eight current responses* (1461 ADC $\approx$ 45 mV) — an "equalize around where we already operate" choice, not one derived from an independent requirement. It is not rate-informed: the rate curves (Fig. 7) show no plateau, so there is no counting-efficiency optimum to aim at, and the plastic threshold in the n_TOF PSA is an analysis quantity rather than a hardware trigger level. The geometric mean minimizes the collective HV movement (the suggested steps stay within −120/+117 V of present values, inside the scanned, validated range) and keeps the fleet's occupancies and pile-up behavior close to the operating point already characterized. Setting the absolute scale to something physically motivated — e.g. a plastic MIP at a chosen mV value — is deferred until the liquid-scintillator readout allows a true plastic MIP calibration; the power laws measured here then convert any such target into voltages directly.

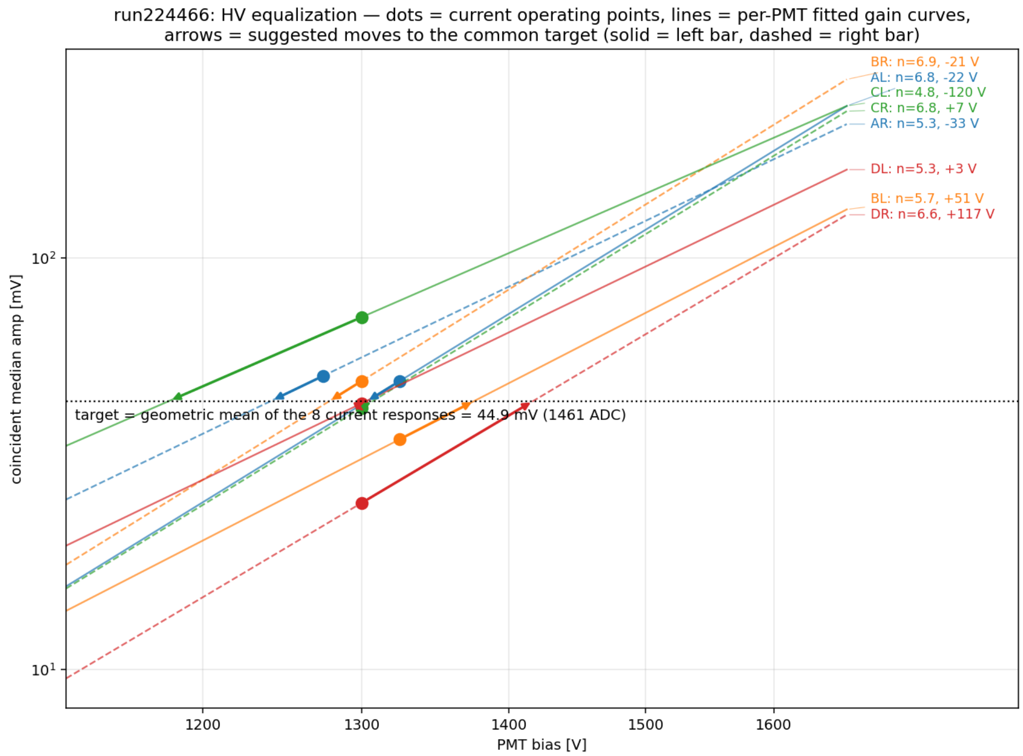

Figure 5: The equalization construction. Dots: current operating points (tagged median from the post-scan window). Lines: per-PMT power laws $G(V) = \mathrm{med}_{\mathrm{now}}(V/V_{\mathrm{now}})^n$. Arrows: suggested moves to the common target (dotted line), the geometric mean of the eight current responses. Right-edge labels give each PMT's index and suggested ΔV.

Table 1: Fitted gain indices and voltage equalization to the fleet geometric-mean tagged median (1461 ADC = 44.9 mV). V_{now} are the operating voltages after the scan; medians from the post-scan window (per-channel ADC→mV from the DAQ full scale, ≈ 0.0307 mV/count).

PMT	V_{now} [V]	median [ADC]	median [mV]	n	V_{sugg} [V]	ΔV [V]
A-L (PSSA1)	1325	1637	50.2	6.83	1303	−22
A-R (PSSA2)	1275	1679	51.7	5.34	1242	−33
B-L (PSSB1)	1325	1178	36.2	5.71	1376	51
B-R (PSSB2)	1300	1637	50.2	6.90	1279	−21
C-L (PSSC1)	1300	2334	71.8	4.83	1180	−120
C-R (PSSC2)	1300	1406	43.2	6.79	1307	7
D-L (PSSD1)	1300	1442	44.3	5.35	1303	3
D-R (PSSD2)	1300	826	25.4	6.59	1417	117

4 Stability of the wall MIP calibration

The scan is also a controlled test of the MIP method itself: if the wall–plastic coincidence selection biased the wall amplitude, the wall MIP peak would move as the plastic gain — and with it the tag rate and its pulse-height mix — changes by a factor of a few. It does not. Figure 6 shows the channel-summed, sideband-subtracted wall MIP peak per arm and per step, normalized to its scan mean: all four arms stay within one amplitude bin (±2.6%) from 1200 V to 1600 V and across the gating boundary. The wall MIP calibration (channel-summed on this run: A ≈ 817 ADC = 25.0 mV, B ≈ 1191 = 36.4 mV, C ≈ 1175 = 35.9 mV, D ≈ 1162 = 35.5 mV) can therefore be used independently of the plastic operating point, including after the equalization of Table 1.

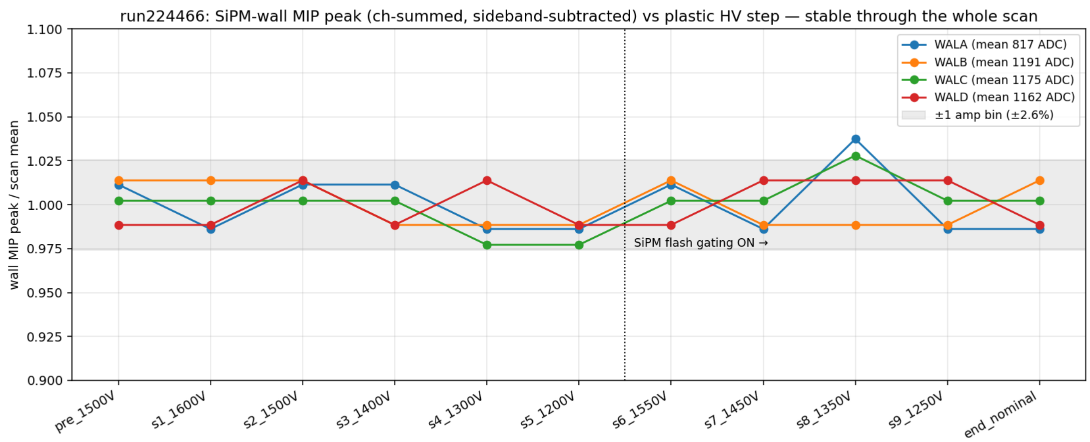

Figure 6: Wall MIP peak (channel-summed, sideband-subtracted) per arm vs. scan step, normalized to the scan mean. Gray band: one histogram bin (±2.6%). Dotted line: SiPM flash-gating turned on.

5 Rates and the flash-gating check

The late-time plastic hit rate per proton rises monotonically with voltage with no plateau up to 1600 V (Fig. 7) — the threshold sits inside the physical spectrum at every voltage, so counting-plateau HV setting is not applicable here; the equalization above is the amplitude-based alternative. The rates are unaffected by the gating state (pass-1 and pass-2 points fall on one curve). In the γ-flash window itself, wall occupancy drops by only ∼4–6% with gating on — much weaker than a blanking of the SiPM readout during the flash would suggest; what the M6.C “SiPM enable” pulse actually diverts should be checked with the hardware experts.

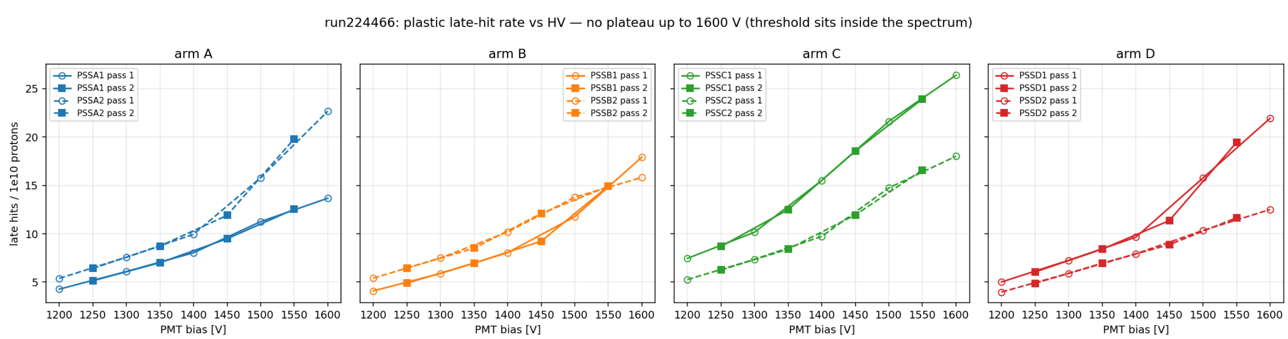

Figure 7: Plastic late-time hit rate per 10^{10} protons vs. bias. Solid: left bars; dashed: right bars; open/filled markers: pass 1/2.

6 A first plastic trigger threshold

If the plastics are to participate in a trigger, the natural question is how low the discriminator can sit. There is no signal/noise valley to aim for: the tagged spectra rise monotonically toward the acquisition cutoff ($\sim$1.5 mV to 2 mV, the n_TOF PSA threshold of $\sim$50 ADC counts — nothing below it is recorded, so lower settings are unexplored territory). The choice is therefore efficiency-driven. Figure 8 scans a *common* threshold on the post-equalization amplitude scale (each PMT's measured spectrum scaled by its equalization factor, so one value serves all eight): the wall-tagged population — our best proxy for real particles — is kept at $\geq$99% on every PMT down to 4.2 mV and $\geq$95% down to 6.3 mV, while the accepted late-time hit rate falls by $\sim$15% (25%) relative to threshold-free.

Unconstrained, the natural choice would be 4 mV to 5 mV (digitizer-equivalent, post-equalization) — safely above the acquisition cutoff and the near-threshold pile-up, $\geq$99% efficient for the tagged population everywhere. **In practice the discriminator hardware has a 10 mV floor**, where the worst PMT keeps 86% of the tagged population (range 86–97% over the eight, at the equalized gains). Two ways forward: (a) accept the 10 mV floor at the present gains — the current plan until a true plastic MIP calibration exists; or (b) raise all gains by a common factor with the per-PMT power laws so that 10 mV sits at today's 99% (95%) point: gain $\times$2.35 ($\times$1.59), i.e. equalized voltages moving to 1408–1614 V (1298–1520 V) — only the $\times$2.35 value for D-R exceeds the validated scan range. Higher gain also multiplies the accepted rate and moves the tagged median to $\sim$105 mV; pile-up and dynamic range should be rechecked before adopting it. Caveats: these are digitizer-equivalent mV (map to hardware discriminator units before setting); the rate axis is the late-TOF in-spill sample (relative, not an absolute trigger rate — the γ-flash region adds its own, much larger, instantaneous rate). These numbers are exported to the DAQ repo (`calibrations/pss_trigger/`).

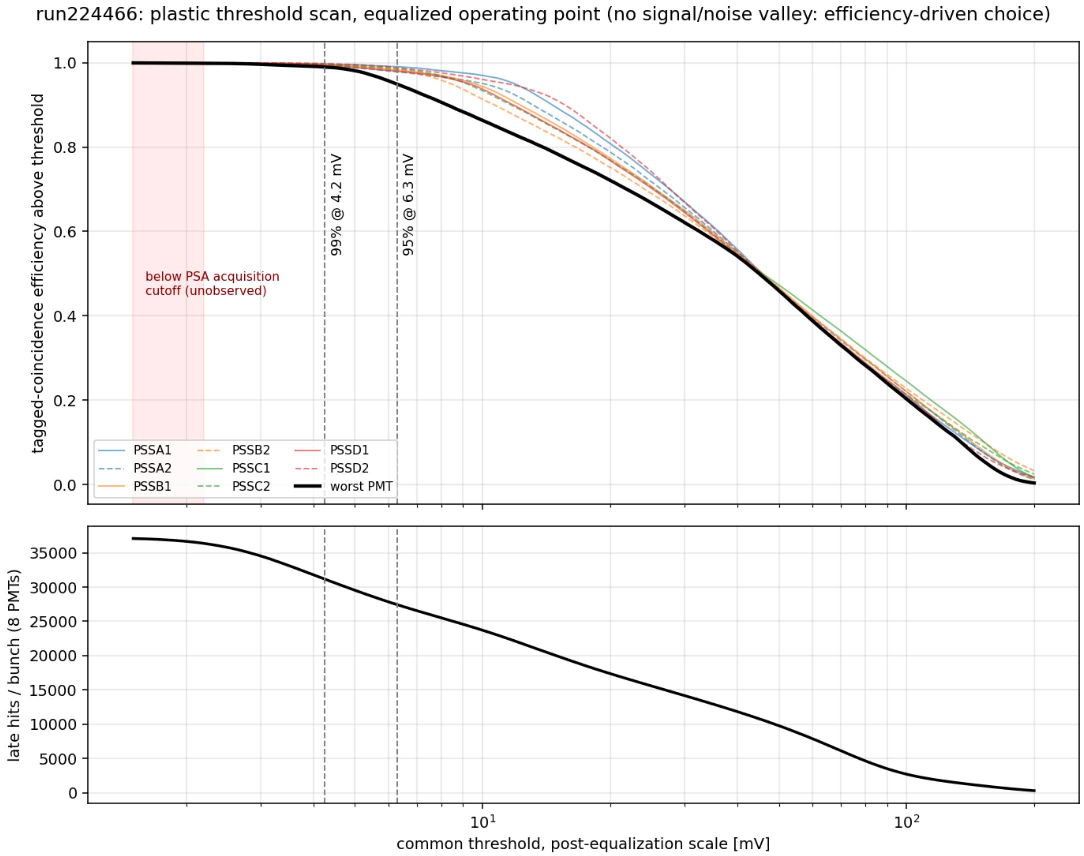

Figure 8: Common-threshold scan at the equalized operating point. Top: wall-tagged (sideband-subtracted) efficiency above threshold per PMT (thin, solid = left / dashed = right bars) and the worst PMT (bold); dashed verticals: 99% and 95% points of the worst PMT. Red band: below the PSA acquisition cutoff (unobserved). Bottom: total accepted late-time hits per bunch (8 PMTs, relative scale).

Reproducibility

HV log and scan handoff: `~/beam_july/scint_hv_scan/2026-07-16_13-32-34_plastic_scan_1/` (pulled from the DAQ machine). Analysis: `mx_july_beam_qa/12_plastic_hv_scan.py` (accumulation, cache `cache/12_hvscan_run224466.npz`) and `12b_hv_scan_plots.py` (figures and Table 1), running on the binary hit cache of the fast read pass (`fastread/`, `hitcache.py`; full-run turnaround ∼3 minutes).